

Hyperparameter Optimisation for Machine Learning Guidelines and Topics

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Practical lab description

Hyperparameter Optimisation (HPO), also known as Hyperparameter Tuning, is the process of searching for a configuration of hyperparameters that yields good performance of a model or a learning algorithm. This process is challenging, and many machine learning algorithms are expensive to run and evaluate. In this lab, you will get hands-on experience with different HPO methods. Starting with both simple grid and random search, you will learn how to implement and analyse HPO from scratch. The lab then introduces more sophisticated model-free and model-based approaches such as Bayesian optimisation. You will also learn how to visualise and analyse the results of experiments.

Practical lab procedure

The lab contains two assignments:

- Assignment 1: implementing common hyperparameter optimisation baselines: random search, successive halving and Bayesian optimisation. For this assignment, you will submit a short report and the code.
- Assignment 2: exploring the state of the art. In this assignment, each group will be assigned a paper from the field of hyperparameter optimisation. You will need to re-implement the paper and perform experiments with the method. For this assignment you will submit a report, code and present your findings.

The lab has several **mandatory** meetings: Meetings from 14:30-16:00 in the seminar room of Theaterstraße 35-39 3rd Floor

- 8.04.26: we will introduce the topic of hyperparameter optimisation and several common methods. We will also cover the basics of experimental setup, visualisation, and analysis of results. We will also give a short introduction to the papers you can choose to implement in the second assignment.
- 6.05.26: discussion on the first assignments, paper assignments for the second assignment.
- 1.07.26: presentations of the second assignment. From 9:00 to 18:00. Schedule to be determined

Additional information on the first assignment is available on Moodle. Office hours by appointment via email are every week from 14:30-16:00.

Reports and presentations

- Each report (for the first and second assignment) is required to follow traditional scientific writing structure. Therefore it needs to have introduction, conclusion and bibliography among other sections.
- At the end of the final report of the second assignment, a statement of contributions is required, which specifies which teammate contributed which parts of the report. In addition, you must disclose whether you used generative AI in your report and how.
- It is mandatory to submit a progress report (PDF + code) with explanation of the methodology used, experimental setup and preliminary results. There should be an outline of other sections, including bullet points.
- Both reports and the presentation must use the provided $\text{\LaTeX}$ templates on Moodle.
- For the presentation, keep the time limit (20 min; a little less is OK, a little more is not OK).
- Each group member should participate equally in all parts.

Criteria for successful completion

- Submission of code and report for both assignments, including the progress report for the second assignment.
- 20 minute presentation + 20 minutes discussion
- Meeting all deadlines
- Attendance of all mandatory meetings
- Grading: 20% first assignment, 40% second assignment, 30% presentation and discussion, 10% participation
- To pass the lab, all parts must be passed individually (first assignment, progress report, second assignment, presentation and attendance).
- No usage of AI outside of polishing reports (tools such as Grammarly)

Important Dates

- First assignment: 5 May 2026, 23:59 CEST
- Papers preferences for the second assignment: 5 May 2026, 23:59 CEST
- Second assignment progress report: 27 May 2026, 23:59 CEST
- Final report due: 17 June 2026, 23:59 CEST.
- Presentations due: 1 July 2026 8:00 CEST
- Submission via Moodle

Papers list for the 2nd assignment

- Stefan Falkner, Aaron Klein, and Frank Hutter. “BOHB: Robust and Efficient Hyperparameter Optimization at Scale”. In: *Proceedings of the International Conference on Machine Learning (ICML)*. vol. 80. 2018, pp. 1436–1445
Required to implement: Bayesian optimisation with random forest (calculation of mean and variance), hyperband, sampling configurations for hyperband with BO (as done in the BOHB algorithm)
- Tree structured Parzen Estimator from James Bergstra et al. “Algorithms for Hyper-Parameter Optimization”. In: *Proceedings of the International Conference on Advances in Neural Information Processing Systems (NeurIPS)*. vol. 24. 2011
Required to implement: TPE as a surrogate for SMBO
- Alexander I. Cowen-Rivers et al. “HEBO: An Empirical Study of Assumptions in Bayesian Optimisation”. In: *Journal of Artificial Intelligence Research* 74 (2022), pp. 1269–1349
Required to implement: input and output transformation, MACE acquisition and optimisation with NSGA-II. Gaussian process implementation should use a well-known GP library that can support the transformations (not scikit-learn).
- Noor H. Awad, Neeratyoy Mallik, and Frank Hutter. “DEHB: Evolutionary Hyperband for Scalable, Robust and Efficient Hyperparameter Optimization”. In: *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence (IJCAI)*. ed. by Zhi-Hua Zhou. 2021, pp. 2147–2153
Required to implement: hyperband, differential evolution and its application as described in the paper.
- Anastasia Makarova et al. “Automatic Termination for Hyperparameter Optimization”. In: *International Conference on Automated Machine Learning, AutoML 2022, 25-27 July 2022, Johns Hopkins University, Baltimore, MD, USA*. vol. 188. Proceedings of Machine Learning Research. 2022, pp. 7/1–21 *Required to implement:* Bayesian optimisation with Gaussian processes, acquisition function and termination criterion.
- Matthias Feurer, Benjamin Letham, and Eytan Bakshy. “Scalable Meta-Learning for Bayesian Optimization”. In: *CoRR* abs/1802.02219 (2018)
Required to implement: Meta learning, ensembling, acquisition function
- Noor H. Awad et al. “Squirrel: A Switching Hyperparameter Optimizer”. In: *CoRR* abs/2012.08180 (2020)
Required to implement: Bayesian optimisation, differential evolution, switching between the two methods. Warmstarting is not required.
- Masashi Shibata et al. *Team Optuna Developers’ Method for Black-Box Optimization Challenge 2020*
Required to implement: Thompson sampling, trust regions, handling of categorical hyperparameters.

- Dimitrios Stamoulis et al. “Hyperpower: Power-and memory-constrained hyper-parameter optimization for neural networks”. In: *Proceedings of the Interantional Conference on Design, Automation & Test in Europe Conference & Exhibition (DATE)*. 2018, pp. 19–24
Required to implement: Predictive Models for Energy, Both acquisition function adaptations
- Antonio Candelieri, Andrea Ponti, and Francesco Archetti. “Fair and green hyperparameter optimization via multi-objective and multiple information source Bayesian optimization”. In: *Machine Learning* 113.5 (2024), pp. 2701–2731**
Required to implement: Implement FANG-HPO
- Michael Emmerich, Nicola Beume, and Boris Naujoks. “An EMO Algorithm Using the Hypervolume Measure as Selection Criterion”. In: *Evolutionary Multi-Criterion Optimization*. Vol. 3410. 2005, pp. 62–76
Required to implement: Implement SMS-EMOA *Not required to implement:* Hypervolume (S-Metric)
- Frank Hutter, Holger H Hoos, and Kevin Leyton-Brown. “Sequential model-based optimization for general algorithm configuration”. In: *Proceedings of the Fifth International Conference on Learning and intelligent optimization (LION)*. 2011, pp. 507–523
Required to implement: Random Forest as surrogate for Bayesian optimisation (calculation of mean and variance), local search as implemented in SMAC
- Martin Wistuba, Nicolas Schilling, and Lars Schmidt-Thieme. “Learning hyperparameter optimization initializations”. In: *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics (DSAA)*. Oct. 2015, pp. 1–10. (Visited on 11/12/2024)
Required to implement: Given a pool of model configurations, implement Learning Initialisation and adaptive learning Initialisation described in the paper

Additional remarks

- In all communication, we use the names as they appear on Moodle. If you would like us to use a different name, please let us know as soon as possible.
- It is possible to de-register from the seminar up to a week after the kickoff meeting. However, if you decide to stop participating after this date, you will receive a failed grade (5.0).
- You are not allowed to modify the provided $\LaTeX$ templates! This especially means modifying the page sizes, margins and font type and size.
- For any question or problem, please contact us: Hadar Shavit shavit@aim.rwth-aachen.de and Nick Kocher kocher@aim.rwth-aachen.de. In case a meeting is needed to clarify questions related to the seminar, we would be happy to schedule one.